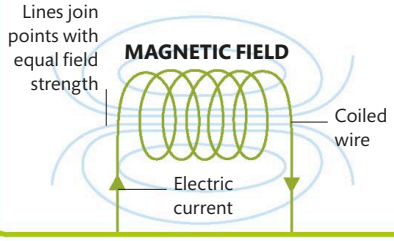
MRI scanner

Magnetic resonance imaging (MRI) is a technique in which a powerful magnetic field and radio waves are used to produce detailed images of the body's internal structures.

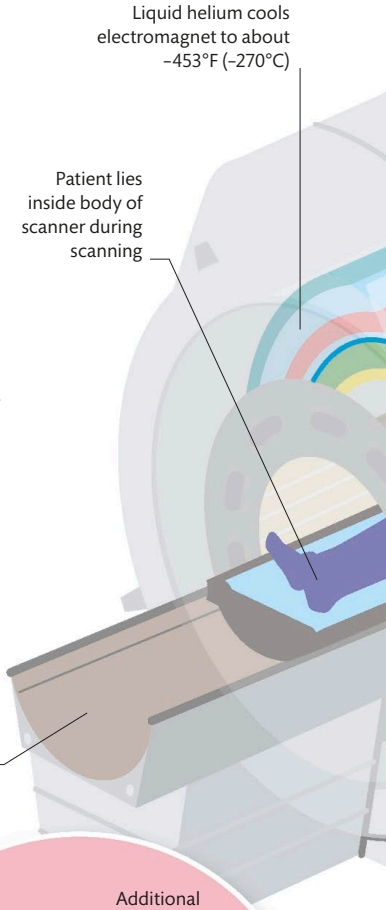
ELECTROMAGNETS

Passing an electric current through a wire creates a magnetic field, turning the wire into an electromagnet. The stronger the current, the stronger the magnetic field. The superconducting electromagnet in an MRI scanner is supercooled with liquid helium to give almost no electrical resistance, allowing very high currents to flow through the electromagnet and produce an extremely strong magnetic field.

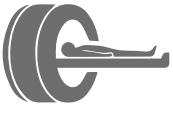


How an MRI scanner works

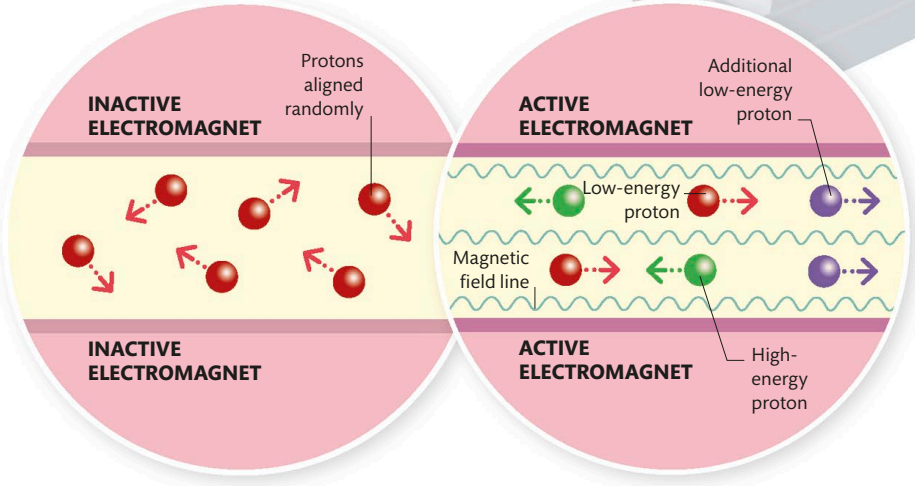
An MRI scanner contains magnets and a radiofrequency coil. A motorized bed moves the patient inside the machine. The main electromagnet generates a very strong magnetic field that makes protons (positively charged particles in atoms) inside body cells align. Gradient magnets alter the field in order to select the specific area of the body to be imaged. The radiofrequency coil emits pulses of radio waves to excite the protons. Radio signals from the protons are then detected by the radiofrequency coil and sent to a computer, which processes the radio-signal data into an image. The MRI image is similar to an X-ray or CT scan (see pp.234–235) but shows more detail, especially in soft tissues.



The scanning process
MRI acts on protons that make up the nuclei of atoms of hydrogen, one of the most abundant elements in the body. It works by making the protons align with a strong magnetic field then exciting them with radio waves and detecting the energy they give off when they return to their previous positions.



THE **ELECTROMAGNET** IN AN **MRI SCANNER** GENERATES A **MAGNETIC FIELD** UP TO **40,000 TIMES** AS STRONG AS **EARTH'S**



1 Normal state of protons
The nucleus of each hydrogen atom consists of a proton. Each proton has a tiny magnetic field, and it spins around the axis of the field. Normally, the protons spin in random directions.

2 Electromagnet turned on
When the electromagnet is on, the protons align along the magnetic field. They may lie in the same direction as the field (a low-energy state) or be counteraligned (a high-energy state). There are slightly more aligned protons than counteraligned ones.